

# Chapter 1: Physical Quantities and Units.



## 1.1) Physical Quantities

All physical quantities consist of a numerical magnitude and a unit.

Eg: length - 50m, Mass - 12 kg, Time - 120 seconds

| Physical Quantity            | Reasonable Estimate |
|------------------------------|---------------------|
| mass of 3 (330ml) coke cans  | 1kg                 |
| mass of medium sized car     | 1000kg              |
| Length of a football field   | 100m                |
| Reaction time of a young man | 0.2s                |



## 1.3) Errors And Uncertainties



Uncertainty is an estimate of the difference between measurement reading and the true value, random & systematic errors lead to uncertainty

**Systematic errors:** Caused by faulty instruments or flaws in method. errors affect accuracy of reading and can be reduced by recalibration

**Random errors:** Caused by fluctuations in instrument reading due to uncontrollable factors. Affects Precision of measurement and to reduce you can take repeat measurements and calculate an average.

**Zero error:** instrument gives non-zero reading when the true reading is zero. Reading must be  $\pm$  the error when recorded.

**Precision:** How close measured values are to each other, if repeated several times and values similar/same, it's precise.

**Accuracy:** How close measured values are to true value, increased by repeating measurements and finding a mean average.

1234 more accurate than 0.123 because more Significant figures, 0.123 is more precise because the last significant figure is in the thousandths place.

### Calculating Uncertainty

**Absolute Uncertainty:** Fixed quantity of uncertainty ( $\pm 0.15$ )

**Fractional Uncertainty:** Uncertainty given as fraction of measurement

**Percentage Uncertainty:** Uncertainty given as Percentage of measurement

Uncertainty (or error) should be recorded to 1 s.f. and the number of decimal points for calculated value is determined by the uncertainty.

## 1.2) SI Units

SI Derived Quantities

SI base units:

Mass (kg)

Length (m)

time (s)

Current (A)

temperature (K)

Area ( $L^2$ ) =  $m^2$

Volume ( $L^3$ ) =  $m^3$

Density ( $\frac{M}{V}$ ) =  $kg m^{-3}$

Velocity ( $\frac{L}{t}$ ) =  $m s^{-1}$

Acceleration ( $\frac{\Delta v}{t}$ ) =  $m s^{-2}$

Momentum ( $m \times v$ ) =  $kg m s^{-1}$

Derived units

SI Prefixes

Tera (T) —  $\times 10^{12}$

Giga (G) —  $\times 10^9$

Mega (M) —  $\times 10^6$

kilo (k) —  $\times 10^3$

Deci (d) —  $\times 10^{-1}$

Centi (c) —  $\times 10^{-2}$

Milli (m) —  $\times 10^{-3}$

Micro ( $\mu$ ) —  $\times 10^{-6}$

Nano (n) —  $\times 10^{-9}$

Pico (p) —  $\times 10^{-12}$

Force ( $\frac{F}{t}$ ) =  $kg m s^{-2}$  (N)

Pressure ( $\frac{F}{A}$ ) =  $kg m^{-1} s^{-2}$  (Pa)

Energy ( $F \times d$ ) =  $kg m^2 s^{-2}$  (J)

Power ( $\frac{E}{t}$ ) =  $kg m^2 s^{-3}$  (W)

Frequency ( $\frac{1}{t}$ ) =  $s^{-1}$  (Hz)

Charge ( $I \times t$ ) = A s (C)

Potential difference ( $\frac{E}{Q}$ ) =  $kg m^2 s^{-3} A^{-1}$  (V)

Resistance ( $\frac{V}{I}$ ) =  $kg m^2 s^{-3} A^{-2}$  ( $\Omega$ )

**Homogeneity:** Combined units must be same

on both sides. Eg:  $S = ut + \frac{1}{2} at^2$

★ all units =  $\frac{m}{m} = m s^{-1} (s) + \frac{1}{2} (m s^{-2} s^2)$

therefore, homogenous =  $m$

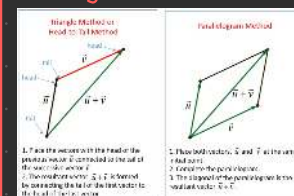
## 1.4) Scalars And Vectors



**Scalar:** quantities that have only magnitude (size) and no direction. Eg: Distance, Speed, Time, Temp.

**Vector:** quantities that have both magnitude and also direction. Eg: Displacement, Velocity, Force, Acceleration.

### Adding And Subtracting Coplanar Vectors

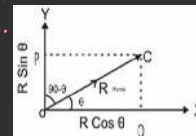


Coplanar vectors are vectors that lie in the same plane.

### Representing a vector as two perpendicular Components

any vector can be broken down into two perpendicular components, one along x-axis & one along y-axis.

R is the magnitude of vector,  $\theta$  is angle made with horizontal axis.



# Chapter 2: Kinematics



## Definitions And Concepts

**Distance:** total length covered irrespective of the direction of motion

**Displacement:** Distance moved in a certain direction

**Speed:** Distance travelled Per unit time

**Velocity:** rate of change of displacement / displacement Per unit time.

**Acceleration:** rate of change of velocity.

### Graph Concepts

**Displacement:** Area under velocity-time graph

**Velocity:** Gradient of Displacement-time graph

**Acceleration:** Gradient of Velocity-time graph



## Equations of Motion

$$v = u + at$$

where

$$s = \frac{1}{2}(u + v)t$$

S: displacement

$$v^2 = u^2 + 2as$$

v: final velocity

$$s = ut + \frac{1}{2}at^2$$

u: initial velocity

t: time taken

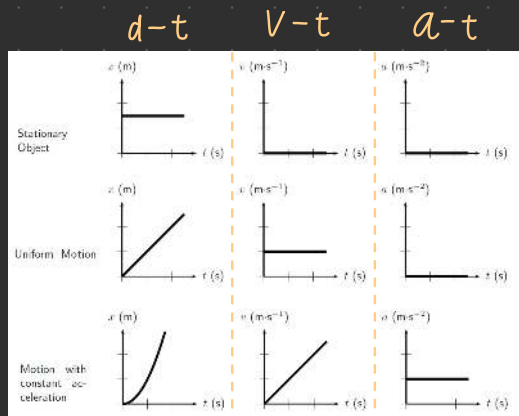
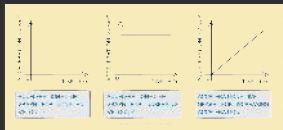
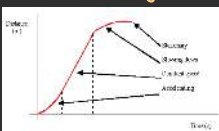
a: acceleration

# Graphical Representation of Complex Motion.

## distance-time graph.

## Velocity-time graph.

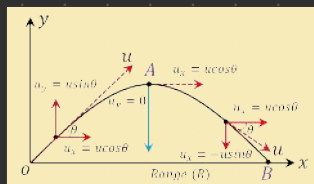
## Acceleration-time graph.



## Motion in two directions:

Uniform Velocity in one direction and uniform velocity in a perpendicular direction creates **projectile motion**.

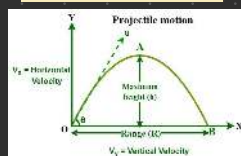
This results in a parabolic trajectory. Combining horizontal velocity with vertical acceleration due to gravity.



$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$



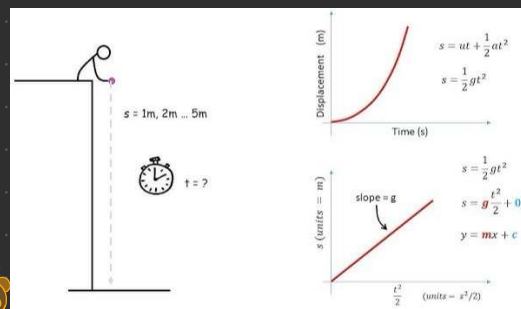
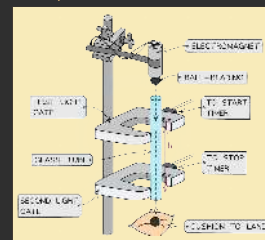
# Experimental Methods.

## Experiment to determine free fall acceleration:

drop an object from known height and measure the time it takes to fall.

use  $S = \frac{1}{2}gt^2$  to calculate g, where S is height, and t is time.

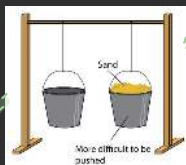
Apparatus: A stopwatch, a ruler, a small object (to drop).



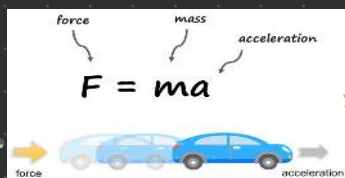
# Chapter 3: Dynamics.

## 3.1) Momentum And Newton's Laws of Motion.

Mass is a measure of an object's inertia, or its resistance to changes in motion. (its tendency to remain at rest or move at constant velocity unless acted upon by a force). The more mass an object has, the more force is required to change its motion.

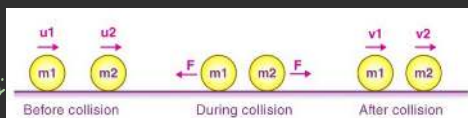


$F = Ma$ , where F: Force, M: mass, A: acceleration. The acceleration of an object is directly proportional to the force applied to it and inversely proportional to its mass. Both the acceleration and resultant force are always in the same direction.



## Linear momentum.

$p = m \times v$  where, p: momentum, m: mass, v: velocity. Momentum is a vector quantity, it is conserved in an isolated system when no external forces are acting.



Force (F) can also be defined as the rate of change of momentum.  $F = \frac{dp}{dt}$ , if the mass is constant, this reduces to  $F = ma$ .

## WEIGHT AND GRAVITATIONAL FIELD.

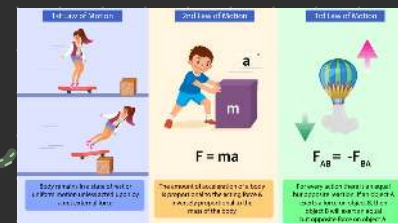
Weight is the force exerted on an object by gravity. it depends on mass & acceleration due to gravity:  $W = mg$ . Weight varies according to g and earth's g is 9.81N/kg. W: weight (N), m: mass (kg), g: gravitational field strength (m/s<sup>2</sup>).

## Newton's Laws of Motion.

#1: First Law (Law of Inertia) an object remains at rest or moves at constant velocity unless an external force acts on it.

#2: Second Law ( $F = ma$ ) the mass of an object is directly proportional to the net force acting on it, directly to mass.

#3: Third Law (Action and Reaction) For every action there is an equal and opposite reaction.



## 3.2) Non-Uniform Motion.

Frictional Forces: Resistive Force that opposes the relative motion of two forces in contact. Types of Friction: Static (Prevent motion between surfaces) and kinetic friction (opposes motion when objects already moving).

Viscous/ Drag Forces: Oppose motion of objects moving through a fluid (air or liquid). Air resistance is a type of drag force.

As the speed of an object increases the drag forces acting on it also increase, providing greater resistance to faster-moving objects.

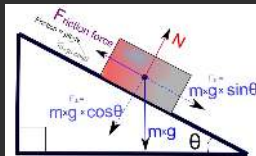
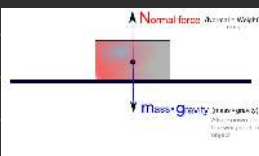
## 3.3) Linear Momentum & Its Conservation.

### Conservation of Momentum

The total momentum of a closed system remains constant unless acted upon by an external force.

Momentum is the product of mass and velocity ( $p = mv$ )

Elastic and Inelastic Collisions



### Motion in Gravitational Field with Air resistance

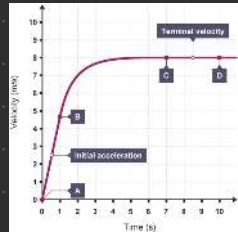
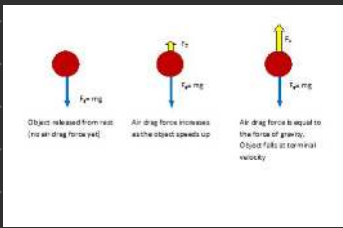
Objects falling in a uniform gravitational field (eg: on earth) are influenced by both gravity pulling them downward and as air resistance grows, it slows the rate of acceleration.

### Terminal Velocity

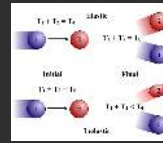
When the force of air resistance becomes equal to the gravitational force, the object stops accelerating.

The object then falls at a constant speed, called Terminal Velocity.

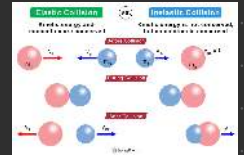
At Terminal Velocity, the net force on the object is zero because the downward force (gravity) is balanced by the upward force (Air resistance).



Elastic collisions: Momentum and Kinetic energy are both conserved.  
Inelastic Collisions: Momentum is conserved, KE is not fully conserved (some lost to heat, sound etc).



where  
 $T_1$ : KE of object 1  
 $T_2$ : KE of object 2  
 $T_0$ : total KE of system



### Relative Speed in Elastic collisions

In an elastic collision, the relative speed of approach between two objects is equal to relative speed of separation after the collision.

$$u_1 - u_2 = v_2 - v_1$$



### Kinetic Energy Changes

While momentum is always conserved, the kinetic energy of a system may change during collisions, especially in inelastic collisions where energy is transformed into other forms.

$$\begin{aligned} \text{PS: Kinetic Energy} &= \frac{1}{2}mv^2 \\ \Delta KE &= \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 \\ &= \frac{1}{2}m(v_2 - v_1)^2 \end{aligned}$$

Example:

A particle of mass 15g was projected at 18 m/s above a horizontal plane. It decelerated uniformly at 2 m/s<sup>2</sup>. Find the change in the kinetic energy in the first 4 seconds of motion.

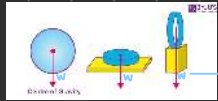
$\Delta KE = \frac{1}{2}m(v_f^2 - v_i^2)$   
 $\Delta KE = \frac{1}{2}(0.015)(0^2 - 18^2)$   
 $\Delta KE = -0.243 \text{ J}$   
 $\Delta KE = -243 \text{ mJ}$

# Chapter 4: Forces, Density and Pressure



## 4.1) Turning Effect of forces

**Centre of gravity:** The point where the entire weight of an object can be considered to act upon.

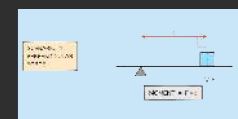


Weight / mass  $\times$  gravitational field strength.

**Moment of a force:**

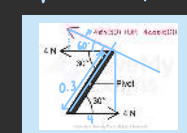
Turning effect of a force about a pivot

Moment = Force  $\times$  distance from pivot [Unit = Nm]

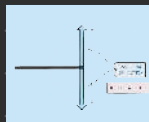


**Couple:** A pair of equal and opposite forces acting on object to produce rotation without translation.

Torque of couple = Force  $\times$  Perpendicular distance between forces.



find force and its PERPENDICULAR distance.

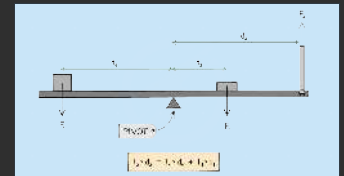


## 4.2) Equilibrium of forces

### Principle of Moments

For an object in equilibrium, the sum of all clockwise moments will be equal to the sum of all anticlockwise moments.

Example



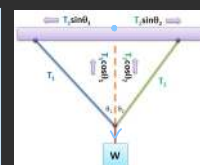
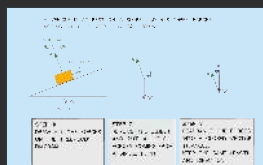
### Conditions for Equilibrium

#1: No resultant forces on objects

#2: No resultant torque (force that cause object to ROTATE).

### Vector triangle

A Vector triangle can be used to represent three coplanar forces in equilibrium.



if objects are in equilibrium,  
 $T_1 \cos \theta + T_2 \cos \theta = W$  — ①  
 $T_2 \sin \theta - T_1 \sin \theta = 0$  — ②

## 4.3) Density and Pressure

### Density

mass per unit volume  $\left[ \rho = \frac{m}{V} \right]$  where:  $\rho$  is density  
 $m$  is mass  
Unit:  $\text{kg m}^{-3}$   $V$  is volume

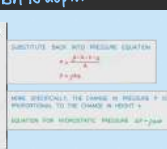
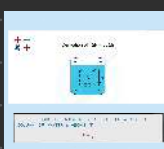
### Pressure

Force Per unit Area  $\left[ p = \frac{F}{A} \right]$  where:  $p$  is pressure  
 $F$  is force  
 $A$  is Area  
Unit:  $\text{N m}^{-2}$  (Pascals)

### Hydrostatic Pressure

Pressure in fluid increases with depth due to the weight of the fluid above.

Formula —  $\Delta P = \rho g \Delta h$  where:  $\rho$  is fluid density  
 $g$  is gravity acceleration  
[Unit: Pascals]  $\Delta h$  is depth



### Upthrust (Buoyancy)

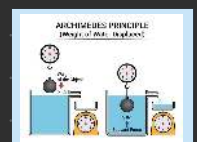
the upward force exerted by fluid on a submerged object due to difference in hydrostatic Pressure.

### Archimedes' Principle

Upthrust  $F = \rho g V$

[Unit: N]

where  $\rho$  is density of fluid  
 $g$  is gravity acceleration  
 $V$  is volume of fluid displaced



# Chapter 5: Work, Energy and Power.



## 5.1) Energy Conservation

### Work

Energy transferred when a force moves an object.

$$W = F \times S \quad \text{where: } W \text{ is work done}$$

[unit: Joules]

F is force

S is displacement in direction of force

### Conservation of Energy

Energy can not be created or destroyed, only transformed from one form to another.

### Efficiency

The ratio of useful energy output to total energy input.

$$\text{Efficiency} = \frac{\text{useful energy out}}{\text{total energy in}} \times 100\%$$

### Power

Rate of doing work or rate of energy transfer.

$$P = \frac{W}{t} \quad \text{or} \quad P = Fv \quad \text{where: } P \text{ is Power, } W \text{ is work done}$$

[unit: Watt/W]

t is time taken, F is force, V is velocity.

## 5.2) Gravitational Potential Energy And Kinetic Energy.

### Gravitational Potential Energy

Energy Stored in object due to height above the ground in a gravitational field

$$\Delta GPE = mg \Delta h \quad \text{where: } m \text{ is mass}$$

[unit: Joule]

g is gravitational acceleration

$\Delta h$  is height / change in height

### Kinetic Energy

The energy an object possesses due to its motion.

$$KE = \frac{1}{2}mv^2 \quad \text{where: } m \text{ is mass of object}$$

V is velocity

$$\Delta KE = \frac{1}{2}m(v_2 - v_1)^2$$

$v_2$  is final velocity

$v_1$  is initial velocity

[unit: Joule]

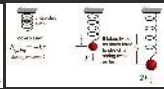
# Chapter 6: Deformation of Solids.



## 6.1) Stress and Strain

Deformation: Caused by tensile forces (Stretching) or compressive forces (Squeezing).

Load: The force applied to an object



Extension: The increase in length of an object under tensile load

Compression: The reduction in length of object under compressive load

Limit of Proportionality: Point beyond which material doesn't follow Hooke's Law.

Hooke's Law:  $F = kx$ , where F is force, x is extension, k is Spring Constant

Stress:  $\sigma = \frac{F}{A}$ , Force per unit Area [Pascals or  $N/m^2$ ]

Strain:  $\epsilon = \frac{x}{L}$ , Extension per original length [no unit]

Young Modulus: ratio of Stress to strain,  $E = \frac{\sigma}{\epsilon}$  [Pascals or  $N/m^2$ ]

Experiment to determine Young Modulus [of metal in wire form]

measure extension of a wire under various loads, then use formula to calculate the stress and strain to calculate the Young Modulus.

## 6.2) Elastic and Plastic Behavior

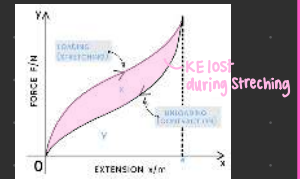
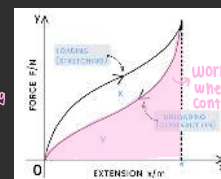
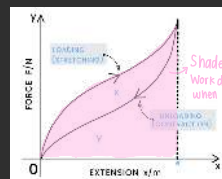
Elastic Deformation: Material returns to original shape when load removed.

Plastic Deformation: Material does not return to its original shape after load removed

Elastic Limit: maximum load that can be applied without causing permanent deformation.

Work done: Area under Force-Extension graph represents work done.

Elastic Potential Energy:  $E_p = \frac{1}{2}Fx$  or  $E_p = \frac{1}{2}kx^2$  for material within limit of Proportionality.



# Chapter 7: Waves

## 7.1) Progressive Waves

Wave motion: Movement of energy through Vibrations, as seen in ropes, springs, and ripple tanks.

### Key terms

Displacement: distance from equilibrium position.

Amplitude: Max displacement from equilibrium position.

Phase difference: measure of how far two points on a wave are.

Period (T): Time for one complete wave cycle.

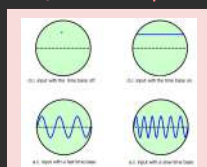
Frequency (f): Number of wave cycles per sec (Hz).

Wavelength ( $\lambda$ ): Distance between successive crest/troughs.

Wave speed (v): Speed at which a wave propagates.

## Cathode Ray Oscilloscope (CRO)

Time base control setting Controls horizontal (x axis), y gain Controls vertical (voltage) scale. It helps measure frequency and amplitude.



## Wave Equation

$$v = f\lambda$$

where

v is wave speed (m/s)

f is frequency ( $s^{-1}$ )

$\lambda$  is wave length (m)

progressive waves carry energy from one point to another.

## Intensity Formulas

$$\text{Intensity} = \frac{\text{Power}}{\text{Area}}$$

$$\text{Intensity} \propto (\text{amplitude})^2 \quad \text{so} \quad I = k a^2$$

$$\text{Intensity} \propto \frac{1}{(\text{radius})^2} \quad \text{so} \quad I = \frac{k}{r^2}$$

## 7.3) Doppler Effect

Doppler effect occurs when a Sound source moves relative to a Stationary observer, the observed frequency differs from the source frequency.

$$f_o = f_s \left( \frac{v}{v \pm v_s} \right)$$

where:  $f_o$  is observed frequency

$f_s$  is source frequency

v is wave speed

$v_s$  is speed of source

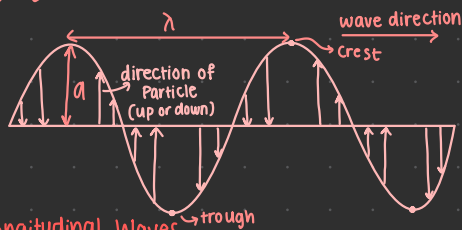
\* Use - when source is moving toward

+ when source is moving away observer.

## 7.2) Transverse & Longitudinal Waves

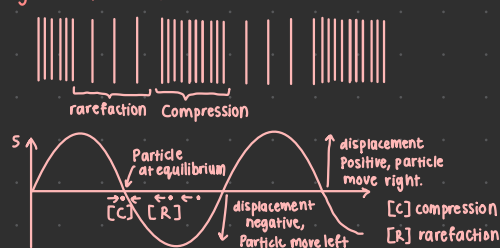
### Transverse Waves

Particles move perpendicular to the wave direction  
Eg: light waves, water waves, seismic-S waves.



### Longitudinal Waves

Particles vibrate perpendicular to wave's direction  
Eg: Sound waves, seismic-P waves.



## 7.4) Electromagnetic Spectrum

Electromagnetic [EM] waves: All EM waves are transverse and travel at speed  $c$  (speed of light in vacuum).

### Wavelength Range

Radio:  $10^3$  m  $\lambda$ , 300 kHz  $f$ , longest  $\lambda$   
microwaves:  $10^{-2}$   $\lambda$ , 30 MHz  $f$   
Infrared:  $10^{-6}$ - $10^{-4}$   $\lambda$ , 3 GHz  $f$   
Visible light:  $10^{-7}$   $\lambda$ , 300 GHz  $f$   
ultraviolet:  $10^{-8}$   $\lambda$ ,  $3 \times 10^{13}$  Hz  $f$   
X-rays:  $10^{-10}$   $\lambda$ ,  $3 \times 10^{15}$  Hz  $f$   
Gamma Rays:  $10^{-12}$   $\lambda$ ,  $3 \times 10^{12}$  Hz  $f$ , longest  $f$ .

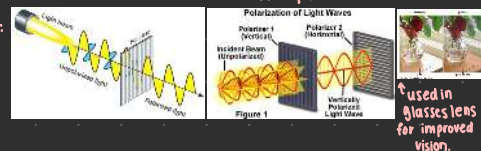
### Visible light

Wavelengths between 400-700 nm are visible to the human eyes.

## 7.5) Polarisation

Polarisation: A property of transverse waves where vibrations occur in one plane

Eg:



### Malus's Law

- describes how the intensity of light changes when it passes through a polarizing filter. It applies to plane polarized light, where the light waves vibrate in a single plane.

$$I = I_0 \cos^2 \theta$$

note: as  $\theta$  increases, less light passes thru and intensity decreases.

where  $I$  is Intensity of transmitted light  
 $I_0$  is Initial Intensity of light before polarization.  
 $\theta$  is angle between direction of light & axis of filter.

note: when  $\theta$  is  $0^\circ$ , all light passes through so  $I = I_0$   
when  $\theta$  is  $90^\circ$ , no light passes through so  $I = 0$

## Chapter 8: Superposition

### Stationary Waves

When two waves meet, the resulting displacement at any point is the sum of the displacements of the individual waves at that point.

Example:

### Stationary waves

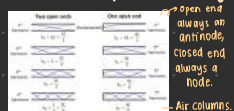
Microwaves: Stationary waves form by reflecting microwaves from a surface, resulting in points of no vibration & max vibration  
Stretched Strings: A string fixed at both ends can show stationary waves by vibrating at specific frequencies (harmonics).

Air Columns: Stationary waves in air columns (like in pipes) form when air vibrates at certain frequencies, with nodes and antinodes in pressure change

Example:



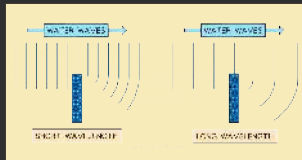
Stretched String harmonics - depends on the frequency of wave.



## Diffraction

Diffraction occurs when waves bend around the edges of an obstacle or pass through a gap and spread out.

Water waves passing through a narrow gap show diffraction. Spreading out more as the gap size approaches the wavelength of the waves.



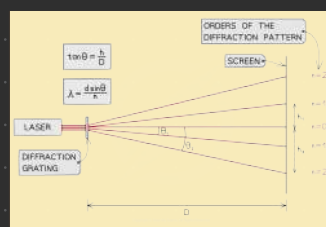
## The diffraction grating

A grating has many slits close together. Light passing through these slits creates an interference pattern.

Diffraction grating formula:  $d \sin \theta = n \lambda$

where -  $d$  is slit separation

- $\theta$  is angle of diffraction
- $n$  is order of maximum
- $\lambda$  is wavelength of light



# Chapter 9: Electricity

## 9.1) Electric Current

### Electric Current:

Electric Current is the flow of Charge Carriers (usually electrons) through a conductor. It is measured in amperes (A), and it flows from higher potential to lower potential.

### Quantisation of Charge

The charge on charge carriers is quantised, meaning that each charge carrier has discrete amount of charge.

The charge on an electron is the smallest unit of charge approximately  $1.6 \times 10^{-19} \text{ C}$ .

Formula:  $Q = It$  where  $Q$ : Charge (Coulombs)  
 $I$ : Current (Amperes)  
 $t$ : time (Seconds)

$I = nAvq$  where  $I$ : Current (A)

$A$ : Cross Sectional Area of conductor ( $\text{m}^2$ )  
 $n$ : number density of charge carriers ( $\text{m}^{-3}$ )  
 $v$ : drift velocity of charge carriers (m/s)  
 $q$ : Charge of each carrier (C)

### Temperature and Resistance of a Filament Lamp

As current increases through a filament lamp its temperature rises, and so does its resistance. This is due to increased vibrations of atoms in the filament, which impede the flow of electrons more as the temperature increases.

### Formula for Resistivity

Resistance of a uniform conductor can be calculated by:

$R = \frac{\rho L}{A}$  where  $R$ : resistance (ohms)  
 $\rho$ : resistivity of material ( $\frac{\text{ohm meters}}{\text{meters}}$ )  
 $L$ : length of conductors (meters)  
 $A$ : Cross sectional area of conductor ( $\text{m}^2$ ).

## 9.2) Potential difference and power.

### Potential Difference:

The potential difference (V) across a component is the energy transferred per unit charge when charge passes through the component. It is measured in volts (V).

$V = \frac{W}{Q}$  where  $V$ : Volts (V)  
 $W$ : energy (J)  
 $Q$ : Charge (C)

### Power

Power is the rate at which energy is transferred. The formulas for Power is:

$P = IV$  where  $P$ : Power (Watts)  
 $V$ : Potential diff (V)  
 $I$ : Current (A)  
 $R$ : resistance (ohms)  
 $P = I^2 R$   
 $P = \frac{V^2}{R}$

## 9.3) Resistance & Resistivity

### Resistance:

Resistance is a measure of how much a component resists the flow of current. It is measured in ohms ( $\Omega$ ) and depends on the material, temperature, and dimensions of the conductor.

### Ohm's Law ( $V=IR$ ):

Ohm's law states that the potential diff across a conductor is directly proportional to the current through it, provided the temperature remains constant.

$V = IR$  where  $V$ : Potential diff (V)  
 $I$ : Current (A)  
 $R$ : resistance (ohms)

### I-V characteristics

**Metallic Conductor**: the current increases linearly with the potential difference at a constant temperature.

**Semiconductor Diode**: A semiconductor diode has a non-linear relationship between current and potential difference. The current increases rapidly after a certain threshold voltage is reached, and the I-V graph is exponential.

**Filament Lamp**: The resistance of a filament lamp increases with temperature as current flows through it. This results in a non-linear I-V graph, where the current increases less sharply as the voltage increases.

# Chapter 10: DC Circuits

## 10.1) Practical Circuits

### Circuit Symbols

|                           |                       |            |
|---------------------------|-----------------------|------------|
| cell:                     | electric bell:        | capacitor: |
| battery:                  | buzzer:               |            |
| Power Supply:             | microphone:           |            |
| A.C. Power Supply:        | Loudspeaker:          |            |
| Junction of conductors:   | motor:                |            |
| Lamp:                     | generator:            |            |
| Fixed Resistor:           | ammeter:              |            |
| Variable Resistor:        | Volt meter:           |            |
| thermistor:               | galvanometer:         |            |
| light-dependant Resistor: | Potentiometer:        |            |
| heater:                   | diode:                |            |
| Switch:                   | Light-emitting diode: |            |
| earth:                    | Oscilloscope:         |            |

### Electromotive Force (e.m.f.):

The e.m.f. of a source is the energy transferred per unit charge in driving the charge around a complete circuit. It is measured in volts (V).

$\text{e.m.f.} = \frac{W}{Q}$  where  $W$ : energy transferred (J)  
 $Q$ : charge (C)

### e.m.f. vs Potential difference (P.d.):

- e.m.f.: total energy supplied by the source to circuit per unit charge.
- P.d.: Energy transferred per unit charge between 2 points in the circuit.

Note: e.m.f. involves energy supplied to the charge, while P.d. involves energy used by the charge.

### Internal Resistance:

The internal resistance of a source reduces the terminal potential difference ( $V_t$ ). The relationship is:

$V_t = \text{e.m.f.} - Ir$  where  $r$  is internal resistance ( $\Omega$ )  
 $I$  is current (A)

Terminal P.d. drops as current increases because of energy lost inside the source.

## 10.2) Kirchhoff's Law

### #1 Law: Current Law

The total current entering a junction is equal to the total leaving it. This is a consequence of the conservation of charge.

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

### #2 Law: Voltage Law

The sum of the e.m.f. in any closed loop is equal to the sum of potential drops in that loop. This is a consequence of the conservation of energy.  $\sum \text{e.m.f.} = \sum \text{P.d.}$

### Resistors in Series

For resistors  $R_1, R_2, \dots, R_n$  in series, the combined resistance is:

$$R_{\text{total}} = R_1 + R_2 + \dots + R_n$$

Derived from Kirchhoff's Second Law.

### Resistors in parallel

For resistors in  $R_1, R_2, \dots, R_n$  in parallel,

## 10.3) Potential Dividers

### Principle of Potential Divider

A potential divider uses two or more resistors in series to produce a desired fraction of the total voltage. The output voltage  $V_{out}$  is:

$$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

### Potentiometer

A potentiometer is a variable resistor used to compare P.d. by adjusting resistance to achieve

a null point (zero current). It allows precise measurement of P.d. between 2 points.

### Galvanometer in null methods

A galvanometer detects zero current in null methods. When it shows no deflection, the circuit is balanced, indicating the P.d. is equal to reference value.

### Thermistors & LDRs in Potential Dividers

Thermistors: As temperature  $\uparrow$ , the resistance of a thermistor  $\downarrow$ , causing the output P.d. to change with temperature.

LDRs: As light intensity  $\uparrow$ , resistance of LDR  $\downarrow$ , altering P.d. based on the light conditions.

the combined Resistance is:

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_n}$$

Derived from Kirchhoff's first Law.

## extra notes:



# Chapter 11: Particle physics

## 11.1) Atoms, nuclei, and radiation

### #1: $\alpha$ -Particle Scattering Experiment:

In Rutherford's experiment, most  $\alpha$ -particles passed through gold foil, but some were deflected at large angles. This concludes that atoms have a small, dense, +vely charged nucleus where most of the mass is concentrated.

### #2: Model of the nuclear Atom:

Atoms consist of a nucleus (protons & neutrons) surrounded by electrons in orbitals. Protons are positively charged, neutrons are neutral, and electrons are negatively charged.

### #3: Nucleon and Proton numbers:

- Proton number (Z): Number of protons in the nucleus (defines the element)
- Nucleon number (A): Total number of protons & neutrons in the nucleus

### #4: Isotopes:

Isotopes are atoms of the same element with the same proton number but different number of neutrons (eg: Carbon-12, carbon-14)

### #5: Nuclide Notation:

Represented as  $A_ZX$  where A is nucleon number, Z is proton number & X is the chemical symbol.

### #6: Conservation in nuclear Processes:

Nucleon number and charge are conserved during nuclear reactions.

### #7: Properties of $\alpha$ , $\beta$ , and $\gamma$ radiation:

- $\alpha$ -particle: Composed of 2 protons, 2 neutrons, charge +2, relatively massive.
- $\beta$ -particle: Electrons ( $\beta^-$ , charge -1) or positrons ( $\beta^+$ , charge +1), small mass.
- $\gamma$ -rays: High-energy electromagnetic radiation, no mass, no charge.

### #8: Antiparticles:

Antiparticles have the same mass but opposite charge as their corresponding particles (eg: Positron is antiparticle of electron)

### #9: Anti(Neutrinos) in $\beta$ -decay:

- $\beta^-$ -decay: Produces an electron & an antineutrino
- $\beta^+$ -decay: Produces a positron & a neutrino

### #10: Discrete and Continuous Energies:

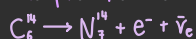
- $\alpha$ -particle: Discrete energy levels
- $\beta$ -particle: Continuous energy due to the emission of (anti)neutrinos sharing the energy.

### #11: Radioactive Decay Equations:

- Example of  $\alpha$ -decay:



- Example of  $\beta$ -decay:



### #12: Unified Atomic Mass Unit (u):

- $1u = \frac{1}{12}$  mass of Carbon-12 atom.  $1u = 1.66 \times 10^{-27} \text{ kg}$

## 11.2) Fundamental Particles

### #1: Quarks

- Fundamental Particles with six types (Flavours):

- up (u):  $+\frac{2}{3}$
- down (d):  $-\frac{1}{3}$
- strange (s):  $-\frac{1}{3}$
- charm (c):  $+\frac{2}{3}$
- top (t):  $+\frac{2}{3}$
- bottom (b):  $-\frac{1}{3}$

- Antiquarks have opposite charges (eg:  $\bar{u} = -\frac{2}{3}$ )

### #2: Protons & neutrons

- Protons: uud (charge +1)
- Neutrons: udd (charge 0)

### #3: Hadrons

Composed of Quarks

Baryons: Made of 3 Quarks (eg: Protons, neutrons)

Mesons: Made of one quark & one antiquark.

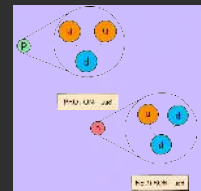
### #4: $\beta$ -decay and Quarks

In  $\beta^-$  decay: a down quark transforms into an up quark, emitting a  $\beta^-$  particle (electron) and an antineutrino.

In  $\beta^+$  decay: an up quark transforms into a down quark, emitting a  $\beta^+$  particle (positron) and a neutrino.

### #5: Leptons

- Fundamental Particles that aren't made of quarks.
- Example: Electrons, neutrinos ( $\nu_e, \bar{\nu}_e$ ).



## extra notes